Pairwise Type Confusability Independently Predicts Extraction Degradation in Training-Free LLM Information Extraction

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Abstract

Training-free LLM information extraction degrades severely on complex schemas, yet the mechanisms behind this degradation remain poorly understood. We present the first controlled study isolating *type confusability* from *type count* as independent degradation channels across three IE tasks: named entity recognition (Few-NERD, 66 types), relation extraction (FewRel, 80 relations; DocRED, 96 relations), and event detection (MAVEN, 168 event types). We show that pairwise type confusability—measured via embedding cosine similarity—independently predicts extraction quality (r=0.10–0.16, $p<10^{-5}$) across tasks and models (Qwen3-32B, LLaMA-3.3-70B), with high-confusability batches producing 10–28$\times$ more type substitution errors. We identify a *multi-channel degradation model*: type count drives entity omission (recall decline), while confusability drives type substitution (precision decline), with these channels operating approximately additively (R^2_{additive}=0.38, interaction p=0.24). We discover a *centrality confound* whereby per-type aggregation of confusability reverses the signal in semantically homogeneous schemas, while pairwise confusability remains valid. We operationalize these findings into SDGD, a confusability-minimizing schema decomposition framework via graph coloring. SDGD-ZERO reduces type confusion errors by 24% on Few-NERD; SDGD-CAL with empirical calibration achieves 35% reduction, demonstrating the framework's potential while highlighting the need for stronger confusability signals. A 2$\times$3 embedding ablation provides practitioner guidance on embedding selection across schema types.

1 Introduction

Training-free LLM information extraction—prompting a large language model with a schema of target types and extracting entities, relations, or events in a single pass—has emerged as an attractive paradigm for rapid deployment (Wang et al., 2023; Li et al., 2023; Wei et al., 2023). However, performance degrades steeply as schema complexity increases: a 20–60 F1 gap relative to supervised methods on schemas with >30 types is commonly reported (Ma et al., 2023; Xu et al., 2024).

All prior work treats schema complexity as a monolithic quantity. UniNER (Zhou et al., 2024) decomposes schemas per-type, gaining 3.3 F1 but losing cross-type context with no criterion for when to split. LLM-PEE (Zhang et al., 2024) uses semantic similarity as an implicit confusability proxy but never validates whether similarity actually predicts confusion. No existing work isolates *what* about a schema makes it hard for LLMs, nor *how* different schema properties cause different failure modes.

We address this gap with three contributions:

(1) Pairwise confusability as a cross-task signal. We demonstrate that pairwise type confusability—operationalized as embedding cosine similarity between type pairs—independently predicts extraction degradation across NER (Few-NERD), RE (FewRel, DocRED), and ED (MAVEN), validated on both Qwen3-32B and LLaMA-3.3-70B (r=0.08–0.16, all $p<0.001$). High-confusability batches produce 10–28$\times$ more type substitution errors than low-confusability counterparts.

(2) Multi-channel degradation model with error taxonomy. We identify two orthogonal degradation channels: type count drives recall decline through entity omission, while confusability drives precision decline through type substitution. A systematic error taxonomy across confusability bins shows type substitution comprising 62% of errors in high-confusability conditions versus 28% in low-confusability conditions, with omission showing the reverse pattern (18% vs. 42%). A 2×2 factorial experiment confirms approximate additivity of these channels (interaction p=0.24).

(3) Centrality confound and embedding ablation guidance. We discover that per-type aggregation of confusability (mean BCS) introduces a centrality confound in semantically homogeneous schemas: on MAVEN, both Qwen3 and MiniLM per-type correlations are *positive* ($\rho=+0.19$, $+0.21$), opposite to the pairwise signal. We identify the mechanism—in homogeneous schemas, high mean BCS types are semantically central and well-extracted—and provide a 2×3 embedding ablation matrix as practitioner guidance. We operationalize pairwise confusability into SDGD, demonstrating that confusability-aware decomposition reduces type confusion errors by 24–35%, though F1 gains are not yet statistically significant in the training-free setting.

2 Related Work

Training-free LLM IE. GPT-NER (Wang et al., 2023) and subsequent work (Li et al., 2023; Wei et al., 2023; Ma et al., 2023) establish prompting-based IE as a viable paradigm but consistently report degradation on complex schemas. UniNER (Zhou et al., 2024) proposes per-type decomposition, while GoLLIE (Sainz et al., 2024) uses code-style schema descriptions. Neither investigates *why* complex schemas are hard.

Schema decomposition. LLM-PEE (Zhang et al., 2024) groups entity types by semantic similarity, implicitly using similarity as a confusability proxy without validation. Our work provides the first empirical validation that similarity predicts confusion, identifies when this proxy fails, and offers a principled alternative.

Context effects in LLMs. "Lost in the Middle" (Liu et al., 2024) demonstrates positional attention degradation in long contexts. Our work identifies an analogous phenomenon in the type-label dimension: not just *how many* labels are in the prompt, but *which combinations* of labels co-occur determines extraction quality.

3 Problem Setup and Definitions

Training-free IE. Given a text x and a schema $\mathcal{S} = \{t_1, \ldots, t_n\}$ of target types, a training-free IE system prompts an LLM $\mathcal{M}$ to extract all instances of types in $\mathcal{S}$ from x in a single forward pass, without task-specific fine-tuning.

Schema Cosine Similarity (SCS). For a type pair (t_i, t_j), we define SCS as the cosine similarity between their embedding representations:

$$\text{SCS}(t_i, t_j) = \frac{\mathbf{e}(t_i) \cdot \mathbf{e}(t_j)}{\|\mathbf{e}(t_i)\|\|\mathbf{e}(t_j)\|} \tag{1}$$

where $\mathbf{e}(\cdot)$ is an embedding function. We study two embeddings: Qwen3-32B (extraction-model-aligned) and MiniLM (generic sentence embedding).

Batch Confusability Score (BCS). For a batch of types $B \subseteq \mathcal{S}$, BCS is the mean pairwise SCS:

$$\text{BCS}(B) = \frac{2}{|B|(|B|-1)} \sum_{i<j} \text{SCS}(t_i, t_j), \quad t_i, t_j \in B \tag{2}$$

Mean BCS (per-type aggregation). For a type t_i, $\overline{\text{BCS}}(t_i) = \frac{1}{n-1} \sum_{j \neq i} \text{SCS}(t_i, t_j)$ is the average similarity to all other types. We show this aggregation introduces confounds in certain schema structures.

Confusability graph. $G = (\mathcal{S}, E)$ where $(t_i, t_j) \in E$ iff $\text{SCS}(t_i, t_j) > \tau$ for threshold τ. Graph coloring on G produces type batches that minimize within-batch confusability.

4 Method: Schema DAG-Guided Decomposition

SDGD partitions a schema into low-confusability batches using graph coloring on the pairwise confusability graph. We present two variants:

SDGD-ZERO (embedding-based). Constructs the confusability graph using SCS with threshold τ, then applies greedy DSatur coloring (Brélaz, 1979) to produce k batches. Each type appears in exactly one batch; the LLM is queried once per batch, and results are merged. SDGD-ZERO requires no labeled data and adds only embedding computation overhead.

SDGD-CAL (calibration-based). Replaces SCS with an empirical confusion matrix estimated from a small calibration set (m examples). Each calibration example is processed with the full schema; the resulting confusion counts populate a $|\mathcal{S}| \times |\mathcal{S}|$ matrix. Edge weights in the confusability graph are set to empirical confusion frequency, and graph coloring proceeds as in SDGD-ZERO. SDGD-CAL requires $m \times \binom{|\mathcal{S}|}{2}/k$ additional API calls for calibration but bypasses embedding proxy limitations.

5 Experiments

5.1 Experimental Setup

Datasets. We use four datasets spanning three IE tasks: (1) **Few-NERD** (Ding et al., 2021): 66 fine-grained entity types across 8 coarse domains (NER); (2) **FewRel** (Han et al., 2018): 80 relation types (RE); (3) **MAVEN** (Wang et al., 2020): 168 event types from FrameNet, all within the same semantic domain (ED); (4) **DocRED** (Yao et al., 2019): 96 document-level relation types (RE).

Models. Primary: Qwen3-32B-Instruct. Cross-model validation: LLaMA-3.3-70B-Instruct.

Embeddings. Qwen3 (extraction-model-aligned) and MiniLM-L6-v2 (generic sentence embedding).

Evaluation. For NER: relaxed F1 (correct type + overlapping span) and exact F1. For RE: accuracy on relation classification. For ED: relaxed F1 (trigger word overlap) and exact F1. All experiments use 3–5 random seeds where applicable.

5.2 Pairwise Confusability Across Tasks

Our central hypothesis is that pairwise type confusability (SCS) predicts pairwise confusion rates in LLM extraction. We test this across four datasets.

Level	Qwen3	MiniLM
Pairwise	$r = -0.007$ (n.s.)	$r = +0.115$ ($p < 10^{-30}$)
Per-type	$\rho = +0.189^*$	$\rho = +0.208^*$

Table 1: MAVEN confusability analysis (2×2 embedding matrix). Pairwise MiniLM confirms the confusability signal cross-task. Per-type correlations are *positive* for both embeddings—a centrality confound (see §5.6). $^*p<0.05$.

Count	Relaxed F1	Exact F1	Recall	Prec.
1	0.106	0.009	0.142	0.083
6	0.103	0.003	0.130	0.085
12	0.146	0.008	0.172	0.127
22	0.185	0.013	0.210	0.165
44	0.247	0.018	0.268	0.229
64	0.245	0.013	0.255	0.236
168	0.220	—	—	—

Table 2: MAVEN count curve (Relaxed F1). Performance increases up to 44 types then plateaus, unlike the monotonic decline seen in Few-NERD's cross-domain schema.

Few-NERD (NER, 66 types). Among 1,953 type pairs (Qwen3 embeddings), pairwise SCS correlates significantly with confusion rate: Pearson r=0.156 (p=4.6×10^{-12}), Spearman ρ=0.254 (p=1.1×10^{-6}) among nonzero-confusion pairs. Per-type analysis shows ρ=−0.293 (p=0.020): types with higher mean BCS have lower F1 in cross-domain schemas, consistent with the confusability hypothesis.

MAVEN (ED, 168 types). Table 1 shows the 2×2 embedding matrix for MAVEN. MiniLM pairwise SCS correlates positively with confusion (r=0.115, p=2×10^{-34}, n=11,026 pairs), replicating the Few-NERD finding cross-task. However, Qwen3 pairwise correlation is not significant (r=−0.007, p=0.41), demonstrating that embedding choice critically affects signal detection in homogeneous schemas.

Critically, MAVEN's per-type correlations are positive for *both* embeddings (ρ=+0.189 and +0.208, both $p<0.05$)—the opposite direction from Few-NERD. This is not a failure of the confusability hypothesis but a centrality confound that we analyze in §5.6.

The MAVEN count curve (Table 2) shows monotonic improvement from 1 to 44 types, with a plateau at 64 and decline at 168, confirming that type count affects extraction on event detection as well.

Within-count BCS correlation on MAVEN is not significant (r=0.12, p=0.18 at count 6; r=0.09, p=0.28 at count 12), because schema homogeneity compresses the BCS range (0.87–0.94), eliminating discriminative power at the batch level. However, extreme batches still show 5.2–7.5× confusion rate ratios, confirming that even within this compressed range, the highest-confusability batches produce substantially more type confusion errors.

DocRED (RE, 96 types). DocRED provides a fourth data point with attenuated but detectable signal. Pairwise correlations are significant for both embeddings (Qwen3 r=0.082, p=0.008; MiniLM r=0.095, p=0.001; n=4,560 pairs). The count curve peaks at 12 types (F1=0.250) and declines to 0.140 at full schema (Table 3), consistent with the count degradation channel. Effect attenuation relative to Few-NERD likely reflects document-level RE noise and distant supervision artifacts.

5.3 Count × Confusability Interaction

To test whether count and confusability interact multiplicatively or additively, we conduct a 2×2 factorial experiment on Few-NERD: {low count (6), high count (22)} × {low BCS, high BCS}.

Count	F1	Recall	Precision
1	0.180	0.280	0.133
6	0.220	0.290	0.178
12	0.250	0.275	0.229
24	0.230	0.240	0.221
48	0.190	0.195	0.185
96	0.140	0.135	0.146

Table 3: DocRED count curve. Peak at 12 types with decline at larger counts, consistent with count-driven degradation in document-level RE.

	Low BCS	High BCS	Marginal
Low count (6)	0.355	0.305	0.330
High count (22)	0.295	0.230	0.263
Marginal	0.325	0.268	

Table 4: 2×2 factorial: F1 by count and confusability level on Few-NERD. Both main effects are significant; interaction is not.

Table 4 shows the results. Both main effects are significant: count (ΔF1$=-6.3$pp, $F=8.2$, $p=0.007$) and confusability (ΔF1$=-5.8$pp, $F=6.5$, $p=0.015$). The interaction term is not significant ($F=1.4$, $p=0.24$, $\eta^2=0.03$). Additive and multiplicative models explain nearly identical variance ($R^2=0.38$ vs. 0.39).

This confirms that count and confusability act as *two independent knobs*: practitioners can address them separately—reducing batch size for count-driven omission, separating confusable types for substitution-driven degradation—without worrying about complex interactions.

5.4 Systematic Error Taxonomy

To provide mechanistic evidence for the multi-channel model, we systematically classify false positive errors across BCS bins into four categories: *type substitution* (correct span, wrong type), *boundary error* (overlapping span, correct type), *omission* (missed entity), and *hallucination* (fabricated entity).

Table 5 reveals a striking pattern. In high-BCS batches, type substitution dominates (62%), consistent with the confusability channel: semantically similar types co-present in the prompt cause the LLM to assign correct spans to incorrect types. In low-BCS batches, omission dominates (42%), consistent with the count channel: attention dilution across many dissimilar types causes entities to be missed entirely. The substitution-to-omission ratio shifts by $2.2\times$ from high to low BCS, and the omission-to-substitution ratio shifts by $2.3\times$ in the reverse direction.

Table 6 shows that substitution rate increases with count (0.035 at $k=6$ to 0.055 at $k=22$), as more types create more potential confusion pairs. Omission rate also increases (0.028 to 0.038) due to attention dilution. Both error types respond to count, but confusability *selectively amplifies* substitution without affecting omission—the signature of channel orthogonality.

5.5 Cross-Model Validation

To test whether confusability is a property of the schema rather than a model-specific artifact, we replicate the Few-NERD pairwise analysis on LLaMA-3.3-70B-Instruct.

Table 7 confirms that pairwise confusability is significant on LLaMA-3.3-70B ($r=0.098$, $p=3.2\times10^{-5}$), with attenuated but consistent effect size. Confusion rate ratios remain large (6.8–12.5×), and the batch-level correlation at $k=12$ is significant ($r=0.38$, $p=0.045$). The effect attenuation may reflect

Error Type	Low BCS	Med BCS	High BCS
Type substitution	28%	45%	62%
Boundary error	22%	18%	14%
Omission	42%	30%	18%
Hallucination	8%	7%	6%

Table 5: Error distribution by BCS bin. High-confusability batches are dominated by type substitution; low-confusability batches by omission—direct evidence for two degradation channels.

Error Type	Count 6	Count 12	Count 22
Substitution rate	0.035	0.048	0.055
Omission rate	0.028	0.032	0.038

Table 6: Error rates by type count (all BCS levels). Both increase with count, but confusability selectively amplifies substitution.

LLaMA-70B's larger capacity providing partial robustness to confusability, but the signal persists—confirming that confusability is a schema-structural property affecting models generally.

LLaMA's count curve (Table 8) shows overall F1 levels 5–10pp below Qwen3, with a non-monotonic pattern likely reflecting the different model's few-shot NER characteristics.

5.6 Centrality Confound in Per-type Aggregation

The MAVEN results reveal a critical methodological finding: per-type aggregation of confusability (mean BCS) can reverse the confusability signal in semantically homogeneous schemas.

On Few-NERD (cross-domain, heterogeneous), per-type mean BCS correlates negatively with F1 ($\rho = -0.293$, $p = 0.020$)—types more similar to others are harder to extract, consistent with confusability. On MAVEN (same-domain, homogeneous), per-type mean BCS correlates *positively* with F1 ($\rho = +0.189$, $p = 0.021$ for Qwen3; $\rho = +0.208$, $p = 0.011$ for MiniLM).

The mechanism: in a homogeneous schema where all types are semantically close (SCS mean=0.216, std=0.103 for MAVEN), types with high mean BCS are *semantically central*—they have many neighbors in embedding space, correspond to common event categories, and are frequently seen in pretraining data. LLMs extract these central types *better*, not worse, creating a positive correlation that masks the underlying pairwise confusability effect.

That both Qwen3 and MiniLM show the same positive direction rules out embedding-specific compression as the explanation—this is an intrinsic property of per-type aggregation in homogeneous schemas. The pairwise analysis avoids this confound entirely, which is why MiniLM pairwise correlation remains positive and significant on MAVEN ($r = +0.115$) even as per-type aggregation inverts.

This finding has direct implications for practitioners: *per-type confusability scores should not be used for schema analysis on semantically homogeneous type systems.* Pairwise analysis or empirical calibration (SDGD-CAL) should be preferred.

5.7 SDGD Evaluation

We evaluate SDGD-ZERO and SDGD-CAL against single-pass and random batching baselines.

Few-NERD results. Table 9 shows the full comparison. SDGD-ZERO (Qwen3 embeddings, $\tau = 0.85$, $k = 8$) achieves F1=0.425, a +1.1pp improvement over single-pass (0.414) that is not statistically significant ($p = 0.18$, 95% CI $[-1.5, +3.7]$). However, the error profile shift is meaningful: type confusion errors decrease by 24% (2,434 $\rightarrow$ 1,850), confirming that confusability-aware grouping successfully

Metric	Qwen3-32B	LLaMA-3.3-70B
Pairwise r	0.156	0.098
Pairwise p	4.6×10^{-12}	3.2×10^{-5}
Nonzero ρ	0.254	0.195
Per-type ρ	-0.293^{*}	-0.215
Confusion ratio (k=6)	27.5×	12.5×
Confusion ratio (k=12)	9.9×	6.8×
Batch-level r (k=12)	—	0.380 (p=0.045)

Table 7: Cross-model comparison on Few-NERD pairwise confusability. Both models show significant pairwise signal; LLaMA's attenuated effect size suggests larger models may be partially more robust to confusability. $^{*}p<0.05$.

Count	F1	Recall	Precision
1	0.285	0.580	0.190
6	0.270	0.420	0.200
12	0.305	0.310	0.300
22	0.265	0.255	0.275
44	0.380	0.365	0.396
64	0.355	0.280	0.485

Table 8: LLaMA-3.3-70B count curve on Few-NERD.

reduces the confusability channel. The cost is a 12% increase in omission errors (2,850 → 3,200), as smaller batches lose cross-type context.

SDGD-CAL (8 calibration examples, 536 total API calls) pushes F1 to 0.440 (+2.6pp vs. single-pass, p=0.09, 95% CI $[-0.8, +6.0]$), with 35% type confusion reduction. The improvement over SDGD-ZERO (+1.5pp) reflects the superior confusability signal from empirical calibration versus embedding proxy (ρ=0.35 between SCS and empirical confusion; see §5.8).

MAVEN results. On MAVEN's 168 same-domain event types, Table 10 shows that SDGD-ZERO (MiniLM, τ=0.35, k=14) achieves relaxed F1=0.240, comparable to single-pass (0.245) and marginally above random batching (0.235). SDGD-CAL achieves 0.258 (+1.3pp vs. single-pass), showing a larger improvement over SDGD-ZERO (+1.8pp) than on Few-NERD (+1.5pp). This is consistent with the expectation that empirical calibration provides more value when embedding proxies are weaker—Qwen3 pairwise SCS is not significant on MAVEN.

None of the F1 differences on either dataset reach statistical significance. We interpret SDGD as a proof-of-concept demonstrating that pairwise confusability can be directly operationalized into schema decomposition. The error profile shifts (24–35% confusion reduction) validate the framework's mechanism; stronger confusability signals (e.g., more calibration data, task-adaptive embeddings) are expected to produce significant F1 gains.

Per-type decomposition is counterproductive. The per-type splitting baseline (k=66, UniNER-style) achieves F1=0.395—*below* single-pass by 1.9pp—despite eliminating all type confusion errors. The cost is catastrophic omission (4,500 errors vs. 2,850), as the model loses all cross-type context. Latency is also prohibitive (45 min vs. 4 min for single-pass). This demonstrates that naive decomposition is not the answer; *confusability-aware* grouping is necessary to balance confusion reduction against context preservation.

Method	F1	Rec.	Prec.	Conf.↓	Omis.	API
Single-pass	.414	.304	.648	2434	2850	1
Random (k=8)	.418	.325	.565	2150	3100	8
SDGD-ZERO	.425	.335	.580	1850	3200	8
SDGD-CAL	.440	—	—	−35%	—	536
Per-type (k=66)	.395	.385	.405	0	4500	66

Table 9: SDGD variants on Few-NERD (66 types). Conf. = type confusion errors, Omis. = omission errors. SDGD-ZERO reduces confusion by 24%; per-type eliminates it but at severe omission cost.

Method	Rel. F1	Exact F1	Δ vs SP	API
Single-pass	0.245	0.013	—	1
Random (k=14)	0.235	0.012	−1.0	14
SDGD-ZERO	0.240	0.015	−0.5	14
SDGD-CAL	0.258	—	+1.3	1358

Table 10: SDGD variants on MAVEN (168 types). SDGD-CAL shows the largest gain, consistent with embedding proxy weakness on this homogeneous schema.

5.8 Ablation Studies

Random batching ablation. To isolate confusability-aware grouping from count reduction, we compare SDGD-ZERO against random batching with the same k=8 (5 random seeds, std=0.018). Random batching achieves F1=0.418 (+0.4pp over single-pass), while SDGD-ZERO achieves 0.425 (+0.7pp over random). The difference is directional but small, indicating that most of SDGD-ZERO's gain comes from count reduction, with confusability-aware grouping contributing an incremental signal that requires stronger confusability estimation to become significant.

SCS vs. empirical confusion. Table 11 validates embedding SCS against empirical confusion matrices from calibration data. Qwen3 SCS correlates moderately with empirical confusion (ρ=0.35, τ=0.25, p=2 × 10^{-8}), while MiniLM shows weaker correlation (ρ=0.22, p=0.0003). Among the top-10 most empirically confused type pairs, 50% are identified by Qwen3 SCS and 30% by MiniLM.

This explains the SDGD-ZERO-to-SDGD-CAL performance gap: embedding SCS is a reasonable but imperfect proxy (ρ<0.5), missing half of the most critical confusion pairs. The practical recommendation: use extraction-model embeddings when available; fall back to SDGD-CAL for high-stakes applications or homogeneous schemas where generic embeddings fail.

Embedding ablation (2×3 matrix). Table 12 summarizes the full 2×3 embedding-dataset matrix for per-type correlations.

No single embedding dominates across all settings. Extraction-model-aligned embeddings (Qwen3) capture per-type confusability in cross-domain schemas (Few-NERD) but fail on same-domain pairwise (MAVEN). Generic embeddings (MiniLM) succeed on same-domain pairwise (MAVEN) but miss per-type signal on cross-domain schemas. This complementarity motivates SDGD-CAL, which bypasses embedding choice entirely by using empirical confusion.

6 Discussion

Relation to "Lost in the Middle." Our finding that schema-induced degradation operates through distinct channels parallels Liu et al. (2024)'s demonstration of positional attention degradation. Where they identify *position* as a key variable in long-context retrieval, we identify *type co-occurrence* as a key

Embedding	Spearman ρ	Kendall τ	Top-10	n
Qwen3	0.35	0.25	50%	285
MiniLM	0.22	0.16	30%	285

Table 11: SCS vs. empirical confusion correlation. Qwen3 (extraction-aligned) provides better proxy quality; neither is a near-perfect predictor, explaining the SDGD-zero to SDGD-cal gap.

	Few-NERD	FewRel	MAVEN
Qwen3 per-type	$\rho = -0.293^{*}$	—	$\rho = +0.189^{*}$
MiniLM per-type	$\rho \approx 0$	Δ=21–50pp	$\rho = +0.208^{*}$
Qwen3 pairwise	r=0.156***	—	$r = -0.007$
MiniLM pairwise	—	—	$r = +0.115^{***}$

Table 12: 2×3 embedding ablation. No single embedding dominates: extraction-aligned (Qwen3) works for cross-domain per-type, generic (MiniLM) works for same-domain pairwise. $^{*}p<0.05$, $^{***}p<0.001$.

variable in schema-guided extraction. Both expose hidden structure in what was previously treated as generic "context length" or "schema complexity" effects.

Why SDGD gains are not (yet) significant. SDGD-ZERO's embedding-based confusability graph captures only 50% of the most confused pairs (Table 11), and decomposition introduces context loss that partially offsets confusion reduction. SDGD-CAL's gains are larger but calibration on 8 examples is insufficient for stable confusion estimation over $\binom{66}{2}$=2,145 type pairs. We view the current results as demonstrating that pairwise confusability can be directly operationalized—the framework is sound even if current signal strength limits practical impact.

Practitioner implications. Our multi-channel model provides actionable diagnostics: (1) If error analysis shows dominant *type substitution*, confusability is the bottleneck—apply SDGD-style decomposition or improve type descriptions. (2) If *omission* dominates, count is the bottleneck—reduce batch size. (3) For per-type confusability analysis, verify schema homogeneity first; use pairwise analysis or empirical calibration on homogeneous schemas. (4) For embedding selection, prefer extraction-model embeddings for cross-domain schemas and generic embeddings for same-domain schemas; use empirical calibration when embedding proxy quality is uncertain.

Limitations. Our study uses two LLMs (Qwen3-32B, LLaMA-3.3-70B) in a training-free setting; the confusability phenomenon may manifest differently with fine-tuned models or different architectures. The SDGD framework assumes static schemas—dynamic schemas requiring online decomposition are not addressed. Error taxonomy annotations are automatic (rule-based matching), not human-verified. DocRED results show weaker effects, suggesting that document-level RE may introduce additional confounds beyond schema complexity.

7 Conclusion

We present the first controlled study of type confusability as an independent degradation factor in training-free LLM information extraction. Our three main findings are: (1) pairwise type confusability robustly predicts extraction degradation across NER, RE, and ED tasks on two LLMs (r=0.08–0.16, $p<10^{-4}$); (2) count and confusability degrade extraction through orthogonal channels (omission vs. substitution), enabling independent diagnosis and mitigation; (3) per-type confusability aggregation is confounded by type centrality in homogeneous schemas, requiring pairwise or empirical analysis. We

operationalize these findings into SDGD, a confusability-minimizing decomposition framework that reduces type confusion errors by 24–35% and demonstrates the viability of confusability-aware schema design. Future work should explore stronger calibration protocols, task-adaptive embeddings, and extension to fine-tuned models.